HEAVY METAL CERTIFIED · CERTIFICATION STANDARD August 2026

Ceramicware: HMTc Standard

Karen Pendergrass · · Heavy Metal Tested & Certified, Institute of Contaminant Standards

CITE AS

Pendergrass K. "Ceramicware: HMTc Standard". Heavy Metal Certified, 2026. https://heavymetalcertified.com/standards/-tableware-ceramicware. CC BY 4.0.

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for ceramicware in Food-Contact Consumer Goods and Kitchenware, evaluated on element RELEASE FROM THE GLAZE, VITREOUS ENAMEL OR GLASS NETWORK INTO FOOD OR A FOOD SIMULANT, µg/kg of food, determined by the 4 percent (v/v) acetic acid filled-article or contact-surface extraction convention used for this material family, with decoration within 20 mm of the lip determined separately. This is the glaze-and-glass RELEASE basis; it is deliberately NOT the Reg (EU) 10/2011 plastics-simulant convention, and a plastics specific migration limit is never substituted for a release limit on this family.. No government sets a numeric heavy-metal limit for this product form, so every panel analyte is held under mandatory analytical control or a substantiated not-material determination rather than an invented ceiling.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum for this exact product form expressed as specific element migration into food or a food simulant; a disclosed read-across from the nearest applicable sovereign value in the same basis where the food-contact material family is expressly excluded from or not reached by that instrument; a substantiated not-material determination; or a control. No cell is occurrence-derived, and no cell borrows a value across a basis.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS

Lead (Pb)	Tier 1	control	measured every lot; no human-food maximum
Cadmium (Cd)	Tier 1	control	measured every lot; no human-food maximum
Total arsenic (tAs)	Tier 2	control	measured every lot; no human-food maximum
Inorganic arsenic (iAs)	Tier 1	control	measured every lot; no human-food maximum
Mercury, total (tHg)	Tier 1	control	measured every lot; no human-food maximum
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	control	measured every lot; no human-food maximum
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here

Cadmium (Cd)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot supply a published number without a recorded governance ruling; this is the same treatment already given on the metal rows to Japan Notice 370 and Korea MFDS Notice 2024-29, and it is a jurisdictional gate rather than a judgement about Uruguayan law. Second, the register normalises those rows to a per-MASS figure carrying basis_class as_sold,

Total arsenic (tAs)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot supply a published number without a recorded governance ruling; this is the same treatment already given on the metal rows to Japan Notice 370 and Korea MFDS Notice 2024-29, and it is a jurisdictional gate rather than a judgement about Uruguayan law. Second, the register normalises those rows to a per-MASS figure carrying basis_class as_sold,

Inorganic arsenic (iAs)	control	Arsenic released from a glaze, enamel or glass network is inorganic by construction, so it cannot be bounded below the total-arsenic result, and this row's total-arsenic cell is itself a control with no sovereign ceiling to bound it. A speciation-bounded not-material determination is therefore unavailable and the fail-safe applies. No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot

Mercury, total (tHg)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot supply a published number without a recorded governance ruling; this is the same treatment already given on the metal rows to Japan Notice 370 and Korea MFDS Notice 2024-29, and it is a jurisdictional gate rather than a judgement about Uruguayan law. Second, the register normalises those rows to a per-MASS figure carrying basis_class as_sold,

Methylmercury (MeHg)	n/m	Methylmercury is formed by microbial methylation in aquatic food chains. A glaze, an enamel and a glass network are fired inorganic matrices with no organic carbon and no methylation pathway, and no credible sovereign sets a methylmercury requirement for any food-contact material. Total mercury remains a mandatory control on this row; methylmercury is not material and never substitutes for it.
Nickel (Ni)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot supply a published number without a recorded governance ruling; this is the same treatment already given on the metal rows to

Tin (Sn)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot supply a published number without a recorded governance ruling; this is the same treatment already given on the metal rows to Japan Notice 370 and Korea MFDS Notice 2024-29, and it is a jurisdictional gate rather than a judgement about Uruguayan law. Second, the register normalises those rows to a per-MASS figure carrying basis_class as_sold,

Aluminium (Al)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot supply a published number without a recorded governance ruling; this is the same treatment already given on the metal rows to Japan Notice 370 and Korea MFDS Notice 2024-29, and it is a jurisdictional gate rather than a judgement about Uruguayan law. Second, the register normalises those rows to a per-MASS figure carrying basis_class as_sold,

Total chromium (Cr)	control	No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows are DECLINED on three stated grounds, and not because they are absent. First, Uruguay sits OUTSIDE the HMTc credible-regulator set used for anchoring in this build, so it cannot supply a published number without a recorded governance ruling; this is the same treatment already given on the metal rows to Japan Notice 370 and Korea MFDS Notice 2024-29, and it is a jurisdictional gate rather than a judgement about Uruguayan law. Second, the register normalises those rows to a per-MASS figure carrying basis_class as_sold,

Hexavalent chromium (Cr-VI)	control	Hexavalent chromium is the health-relevant chromium species and it is a live concern on this family: chromate and dichromate raw materials are used in glaze and enamel colourants, and oxidising firing atmospheres can retain chromium in the hexavalent state at the glaze surface. No credible sovereign sets a hexavalent-chromium release ceiling for a glazed, enamelled or glass food-contact article, and this row's total-chromium cell is itself a control, so no published parent bounds the species. The fail-safe applies. No credible sovereign release ceiling for this material family is available to anchor this cell, and the register position is stated exactly rather than by assertion of absence. The regime that governs this material family in the real world is EU Council Directive 84/500/EEC as amended by Directive 2005/31/EC, supported in the United States by the FDA compliance-policy guides on lead in ceramicware and decorated glassware and by the California Proposition 65 tableware consent judgments; none of those instruments is present in data/evidence/regulatory_limits.csv as a row of its own jurisdiction, and each is recorded as a register gap for ingest through the verified regulatory-fetch pathway. What the register DOES hold for this exact article class is named here rather than passed over: four Uruguayan Reglamento Bromatologico Nacional section 12.5.13 rows carrying the same numbers as the EU Category II and Category III release values, namely uy_rbn_pb_ceramic_fillable_objects_4mgkg (lead, 4000 ug/kg) and uy_rbn_cd_ceramic_fillable_objects_0_3mgkg (cadmium, 300 ug/kg), both scoped to food-contact ceramic objects that can be filled, and uy_rbn_pb_cookware_storage_gt3l_1_5mgkg (lead, 1500 ug/kg) and uy_rbn_cd_cookware_storage_gt3l_0_1mgkg (cadmium, 100 ug/kg), both scoped to ceramic cooking utensils, packaging and storage containers with capacity over three litres. Those four rows

4 | Cite this standard

Pendergrass, K. "Ceramicware." In Heavy Metal Tested & Certified (HMTc) Food-Contact Consumer Goods and Kitchenware standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/-tableware-ceramicware. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-tableware-ceramicware.